

SANDANSKI, BulgariaWMO: **157120**Lat: **41.550N**Lon: **23.267E**Elev: **207**StdP: **98.86**Time Zone: **2.00 (EUE)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.6	-4.4	-11.4	1.4	-3.9	-9.4	1.7	-1.7	12.4	3.2	10.4	5.6	1.9	320	0.417

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	36.3	21.8	34.7	21.4	33.2	20.9	23.0	33.8	22.2	32.7	21.5	31.5	2.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.4	14.5	27.4	18.7	13.8	26.7	18.0	13.3	26.2	68.9	33.9	65.9	32.9	63.4	31.6	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.0	8.2	6.5	-9.4	39.0	2.8	1.9	-11.4	40.4	-13.0	41.5	-14.6	42.6	-16.7	44.0		
			-10.0	25.0	2.7	1.6	-11.9	26.1	-13.5	27.1	-15.1	27.9	-17.0	29.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.9	3.2	6.0	9.9	14.3	19.0	23.2	26.0	26.1	21.3	15.4	9.5	4.4	(6)
(7)		DBStd	8.70	4.17	4.17	3.76	3.28	3.10	3.25	2.69	2.65	3.38	3.72	3.95	4.06	(7)
(8)		HDD10.0	623	211	119	48	6	0	0	0	0	0	4	54	179	(8)
(9)		HDD18.3	2041	468	344	261	124	31	3	0	0	11	103	264	433	(9)
(10)		CDD10.0	2418	2	8	45	136	279	396	496	499	338	172	41	6	(10)
(11)		CDD18.3	795	0	0	0	4	52	149	238	241	99	12	0	0	(11)
(12)		CDH23.3	8178	0	0	2	48	451	1508	2637	2588	856	89	1	0	(12)
(13)		CDH26.7	3758	0	0	0	6	110	659	1331	1316	324	12	0	0	(13)
(14)	Wind	WSAvg	2.1	1.9	2.5	2.7	2.3	2.1	2.2	2.3	2.0	2.0	1.7	1.7	1.9	(14)
(15)	Precipitation	PrecAvg	547	40	39	44	50	54	48	39	34	43	52	44	60	(15)
(16)		PrecMax	814	110	127	94	118	114	112	120	90	105	142	96	127	(16)
(17)		PrecMin	272	3	6	7	17	28	10	7	2	6	3	2	1	(17)
(18)		PrecStd	121	25	27	23	25	22	25	25	25	30	31	21	35	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.0	18.9	23.1	27.2	31.5	36.5	38.9	38.3	34.9	28.3	21.4	16.9	(19)
(20)			MCWB	9.8	11.7	13.3	16.5	19.6	22.1	22.1	22.3	20.2	18.3	15.2	13.1	(20)
(21)		2%	DB	12.8	16.4	20.6	24.6	29.4	34.2	36.4	36.4	32.4	25.8	19.0	14.1	(21)
(22)			MCWB	9.0	10.3	12.1	15.2	18.9	21.6	21.8	22.0	19.5	17.2	14.3	11.0	(22)
(23)		5%	DB	10.8	14.3	18.6	22.6	27.7	32.4	34.7	34.7	30.4	23.7	17.2	11.7	(23)
(24)			MCWB	7.8	9.4	11.2	14.2	17.9	21.0	21.4	21.3	18.9	16.2	13.0	9.3	(24)
(25)		10%	DB	9.0	12.4	16.5	20.9	26.0	30.5	33.0	33.1	28.2	21.8	15.3	10.1	(25)
(26)			MCWB	6.8	8.5	10.2	13.2	17.3	20.2	20.9	20.8	18.3	15.5	12.1	8.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	12.5	14.5	17.3	20.9	23.4	23.8	24.1	21.3	19.6	16.0	13.7	(27)
(28)			MADB	13.9	17.1	20.8	25.1	29.6	33.5	34.3	35.2	31.6	26.6	19.9	16.5	(28)
(29)		2%	WB	9.4	11.0	12.8	16.1	19.6	22.4	22.9	22.9	20.3	18.0	14.7	11.4	(29)
(30)			MADB	11.7	15.3	18.8	23.2	27.6	32.4	33.9	34.1	29.9	23.9	17.9	13.6	(30)
(31)		5%	WB	8.2	10.0	11.8	14.9	18.6	21.5	22.1	22.1	19.5	17.1	13.7	9.9	(31)
(32)			MADB	10.2	13.4	17.3	21.2	26.3	31.1	32.7	32.7	28.3	22.1	16.6	11.3	(32)
(33)		10%	WB	6.9	8.9	10.7	13.9	17.7	20.7	21.5	21.3	18.8	16.0	12.6	8.5	(33)
(34)			MADB	8.9	12.1	15.6	19.6	24.8	29.4	31.4	31.4	26.5	20.5	14.9	9.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.9	8.6	9.8	10.5	11.3	12.2	12.8	12.9	11.4	9.9	8.1	6.7	(35)
(36)			MADB	9.3	11.5	13.1	13.6	13.9	14.7	15.1	14.8	14.3	12.7	10.3	8.2	(36)
(37)			MCWBR	6.1	6.6	6.7	6.4	5.9	5.6	5.6	5.3	5.5	6.0	5.9	5.4	(37)
(38)		5% WB	MADB	7.7	10.4	11.3	12.2	12.8	14.0	13.9	13.8	12.6	10.7	8.8	6.4	(38)
(39)			MCWBR	5.3	6.2	6.2	6.0	6.0	5.7	5.4	5.4	5.2	5.5	5.2	4.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.327	0.358	0.403	0.462	0.454	0.447	0.444	0.455	0.432	0.392	0.358	0.333	(40)	
(41)		taud	2.393	2.306	2.206	2.050	2.147	2.209	2.222	2.188	2.224	2.323	2.379	2.388	(41)	
(42)		Ebn at Noon	814	838	837	811	827	833	831	808	798	789	768	771	(42)	
(43)		Edh at Noon	82	104	129	161	150	141	138	139	125	101	83	76	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.74	2.59	3.67	4.69	5.69	6.45	6.74	6.02	4.48	3.12	2.04	1.55	(44)	
(45)		RadStd	0.26	0.34	0.39	0.50	0.35	0.49	0.38	0.40	0.36	0.35	0.19	0.26	(45)	